Hyper-Persona: A Multi-Level Hypergraph framework for Text-Based Personality Prediction

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Abstract

This is the abstract of the paper. It briefly summarizes the problem, methods, and key results or conclusions.

1 Introduction

Personality is the living current of consciousness through which thought, emotion, and habit shape the self. It is not a fixed entity but a dynamic process, an everflowing stream where experience and action continuously redefine who we are [1]. Through observable behavioral cues that reflect consistent patterns of thought and emotion, individuals exhibit enduring dispositions such as sociability, impulsivity, or conscientiousness, which together constitute measurable personality traits [2].

In recent decades, Internet-mediated communication (encompassing instant messaging platforms, social networks, email systems, online forums, and collaborative digital environments) has expanded at an unprecedented rate [3]. These communication forms, driven by technological advancement and accelerated by global disruptions, have become deeply embedded in human social and professional life [4]. Understanding individuals' personalities through their digital interactions offers valuable insights into cognitive tendencies, emotional responses, and behavioral patterns across diverse contexts [5]. Such understanding can enhance interpersonal communication, foster organizational harmony, and improve social and commercial relationships. Consequently, personality prediction holds broad potential in numerous domains, including recommender systems, recruitment and talent acquisition, digital marketing, social network analysis, psychological counseling, human resource management etc. [6].

The study of personality has long been central to psychology, seeking to explain stable behavioral patterns and inter-individual differences. Foundational frameworks such as the Big Five model conceptualize personality as comprising five broad dimensions: Openness (O), Conscientiousness (C), Extraversion (E), Agreeableness(A), and Neuroticism (N) [2]. Traditional assessment tools, notably the Big Five Inventory (BFI) and the Revised NEO Personality Inventory (NEO-PI-R), have provided reliable measures of these traits. However, they are constrained by practical limitations such as reliance on self-reporting, susceptibility to social desirability bias, and limited scalability for large populations [7, 8, 9, 10].

The emergence of large-scale online communication has opened new opportunities to overcome these limitations through Automatic Personality Prediction (APP), the computational assessment of personality traits based on digital behavioral data. Text-based APP, in particular, leverages natural language as a mirror of psychological processes, enabling personality inference from linguistic patterns within human communication [6]. Early computational studies relied heavily on handcrafted lexical and psycholinguistic features (such as word usage, emotional tone, or syntactic complexity) combined with traditional machine learning techniques like Support Vector Machines and logistic regression [11]. Although these approaches established the feasibility of computational personality inference, they were limited by their dependency on shallow linguistic features and weak generalization to diverse domains [6, 12].

With the advent of deep learning, a new era for APP research introduced itself. Neural architectures such as convolutional and recurrent networks enabled models to automatically learn distributed text representations, reducing the need for manual feature engineering [13]. Later, transformer-based encoders (e.g., BERT [14], RoBERTa [15], and XLNet [16]) revolutionized the field by modeling long-range contextual dependencies through self-attention mechanisms [17]. These approaches significantly improved predictive performance, yet they often treated text as a flat sequence, neglecting the hierarchical structure that naturally spans words, sentences, and documents [18, 19]. More recently, graph-based and hybrid models have emerged to address this limitation by encoding semantic and syntactic dependencies explicitly, offering greater structural interpretability [20]. Graph Neural Networks (GNNs) have shown promise in representing inter-word and inter-sentence relations, while transformer-based architectures continue to dominate in contextual understanding [21].

However, both paradigms face challenges: graph-based methods often capture only shallow relationships, while transformer-based encoders frequently overlook explicit hierarchical organization [22]. This has motivated a new line of research in hierarchical text modeling that integrates multi-level representations for more comprehensive and interpretable personality prediction [23, 24, 25]. To this end, we propose *Hyper Persona*, a multi-level hypergraph framework for text-based personality prediction. It models text at three interconnected levels (document, sentence, and word) each containing beneficial features for text-based APP. The document level captures global features, the sentence level encodes local semantics, and the word level contains fine-grained lexical information [26, 27, 28]. *Hyper Persona* follows a four-phase architecture comprising preprocessing, text representation, a feature-selective transformer-based graph encoder, and personality classification. After cleaning and structuring the input text into a hierarchical hypergraph, the model learns relationships across these levels and performs classification based on the integrated representations.

This study sought to address the following research questions:

- RQ1. In what ways can multi-level hypergraph structures improve the representation of semantic and syntactic information for automatic personality prediction?
- RQ2. To what extent can a transformer-based model operating over hypergraph representations capture personality-relevant linguistic signals across multiple levels of abstraction?
- RQ3. How does the proposed hypergraph transformer framework perform in comparison with established text classification baselines in predicting personality traits from textual data?
- RQ4. To what degree does each level of processing contribute to the overall predictive accuracy of the model?

This paper is organized as follows. The Related Works 2 section provides a concise overview of recent developments in text-based automatic personality prediction (APP), categorized into Single-Level 2.1 and Multi-Level 2.2 approaches. The Methodology section 3 outlines the approach used to develop *Hyper Persona*. It first describes the dataset 3.1 employed in this study, followed by an explanation of how *Hyper Persona* represents textual structures and learns their representations for personality prediction. The Results 4 section presents the experimental findings, while the Discussion 4.5 section interprets these results, discusses their implications, and addresses the research questions. Finally, the Conclusion 5 section summarizes the key findings and outlines potential directions for future research.

2 Related Works

Personality research originated in psychology through self-report questionnaires such as the BFI and NEO-PI-R, which, despite their reliability, suffer from limitations including time demands, social desirability bias, and restricted applicability [7, 8, 9, 10]. With the rise of large-scale digital communication, computational approaches to Automatic Personality Prediction (APP) emerged, initially relying on handcrafted linguistic features and statistical learning [6, 12]. These were later surpassed by deep learning models that learned representations directly from raw text, offering significant improvements but largely treating language as flat sequences [18].

Recent efforts have introduced transformers and graph-based methods to capture semantic and syntactic dependencies, thereby offering structural interpretability [23]. However, conventional graphs remain limited in representing the hierarchical organization of text across words, sentences, and documents. At the same time, pre-trained transformers have delivered state-of-the-art results with minimal preprocessing [19], yet they too often ignore explicit structural constraints.

In this work, we divide previous studies into three (word, sentence, and document) regarding the level of text processing, and argue that, despite the central role of textual hierarchy in encoding semantics, insufficient attention has been devoted to modeling personality prediction through multi-level text structures. This gap motivates the distinction between existing single-level text-based approaches 2.1 and emerging multi-level approaches 2.2, the latter aiming to integrate hierarchical organization for more comprehensive and interpretable modeling of personality.

2.1 Single Level

In this section, we review single-level text-based methods, which model personality by treating text as a flat sequence and focusing on only *one level* of the linguistic hierarchy. Generally, these approaches rely on linguistic features [6] or neural encoders [18] that capture either local or global semantic and stylistic patterns. However, they overlook the hierarchical organization of language into documents, sentences, and words, collapsing diverse textual signals into a unified representation and thereby constraining the capacity to capture nuanced multi-level dependencies.

Early works exemplify the dominance of the *document-level* paradigm. Ong et al. [29], aggregated a user's tweets into a single representation treating each user as a text document, and trained Support Vector Machines and XGBoost models on linguistic features. Similarly, Shen et al. [30], collapsed the entire body of a user's emails into a single unit of text (document) for personality inference. In both cases, the choice of document-level integration simplified personality modeling by considering an entire user profile as one block of text, but this came at the expense of overlooking sentence and word-level structures.

Skowron et al. [31], also adopted the document level, where heterogeneous social media modalities and metadata were combined into a single holistic user profile representation. Ramezani et al. [40], integrated all text into a document-level structure and introduced graph attention networks to capture semantic relations among concepts. Although graph reasoning enhanced semantic connections, the reliance on a document-level prevented explicit modeling of hierarchical dependencies across different textual units.

The rise of transformer-based encoders further solidified this trend. Jayaraman et al. [32], ultimately processed user data as a continuous flat sequence making their work a document-level approach. They further employed XLNet, which models bidirectional context through relative positional encoding. Christian et al. [19], ultimately processed user data as a continuous flat sequence which is also considered as a document-level approach, and integrated BERT, RoBERTa, and XLNet with sentiment and emotion features topredict personality from text. Hao Lin and Xiaolei Li [33], integrated contextual and sentiment signals and fused them into one global representation at the document-level. They used these documents to enrich transformer encoders with sentiment knowledge from SenticNet6. Kamalesh and Bharathi [34], introduced a pipeline which operated at the document-level reducing personality modeling to a flat input-output mapping. Then they introduced a Binary-Partitioning Transformer with TF-IGM for discriminative feature extraction.

V Ganesan et al. [35], systematically evaluated GPT-3 for zero-shot personality estimation at document-levl to determine whether a large language model can infer Big Five traits directly from user-generated text without task-specific fine-tuning. They aggregated each participant's social media posts into a single document and prompted GPT-3 with carefully designed instructions to output trait scores.

A smaller subset of research has instead targeted the *sentence-level*. Naz et al. [36], focused on sentence-level signals, byemploying Bi-LSTM and DistilBERT to obtain sentence embeddings which were then aggregated for predicting only the *Openness* trait, restricting the model from capturing inter-sentence dependencies or higher-order structures. Similarly, Leonardi et al. [37], used a multilingual transformer framework to embed Facebook posts at the sentence-level, which addressed cross-lingual polysemy but ultimately reduced each post to a single embedding vector.

Kazameini et al. [38], introduced a word-level personality prediction framework that combines contextual BERT embeddings with an ensemble of bagged Support Vector Machines (SVMs), their approach preserves token-level representations and aggregates them through bootstrap sampling, enabling fine-grained capture of lexical cues. Alsini et al. [39], proposed a word-level approach for predicting the Agreeableness trait, to predict whether a person is emotional or rational, using deep learning on distributed word representations. They extracted Word2Vec and GloVe embeddings from MBTI dataset to encode each token's semantic context and trained LSTM and Bi-LSTM networks to model sequential dependencies among these word vectors.

Overall, single-level approaches demonstrate distinct advantages. Word-level models capture fine-grained lexical cues and psycholinguistic signals, enabling direct exploitation of trait-relevant vocabulary. Sentence-level methods extend this by modeling local syntactic and semantic structures, which is valuable when personality markers emerge from compositional phrases or clause-level patterns. Document-level methods aggregate an individual's entire text profile, yielding stable global representations and simplifying the modeling pipeline while reducing computational cost.

Despite these strengths, each level carries notable limitations. Word-level modeling often ignores the influence of syntax and discourse, leading to fragmented representations and loss of higher-order context. Sentence-level approaches, while sensitive to local dependencies, overlook inter-sentence coherence and broader narrative flow. Document-level aggregation, although holistic, obscures subtle contextual variations and reduces interpretability by flattening diverse linguistic signals. Across all three levels, hierarchical dependencies remain unmodeled, and language is ultimately treated as a homogeneous sequence.

Single-level text-based methods therefore provide a computationally tractable foundation for Automatic Personality Prediction, highlighting lexical, local, or global features in isolation, however none can simultaneously reconcile fine-grained lexical indicators. Single-level approaches, despite their historical impact, are insufficient for capturing the layered linguistic complexity through which personality is expressed.

2.2 Multi-Level

In this section, we review multi-level methods, which extend personality prediction by modeling the hierarchical organization of language across words, sentences, and documents. By explicitly representing relationships between different text levels and linguistic features, these approaches capture richer contextual and structural cues, enabling more nuanced personality inference.

Zhu et al.[25] integrated word and document-level information by constructing a heterogeneous personality graph for each user. They augmented this graph with LIWC-based psycholinguistic knowledge at the word level and employed semi-supervised noise-invariant learning to improve generalization. This design allowed the model to connect fine-grained psycholinguistic cues to broader document-level relational reasoning. In earlier work, Zhu et al.[41] pursued a similar integration of word and document-level signals. They built personality lexicon-based representations, transformed them into heterogeneous word graphs, and embedded them into document-level structures, thereby explicitly linking local lexical cues with global personality expressions.

Johnson and Murty [42] combined word and document-level modeling through a knowledge graph enriched with DBpedia, ontology data, emotion lexicons, and psycholinguistic features. Their method first learned aspect-aware word embeddings, then integrated them into a document-level graph for trait detection. This multi-level pipeline allowed them to leverage psychologically grounded word-level features while still enabling graph-based reasoning across entire documents. Kerz et al. [43] also exploited word- and document-level integration by combining explicit psycholinguistic feature distributions (word-level) with contextualized representations learned through BERT and BLSTM (document level). Their approach demonstrated how handcrafted linguistic features can be synergistically combined with deep encoders to enhance personality prediction.

Salahat et al.[44] similarly targeted word and sentence-level signals by leveraging BERT embeddings for contextualized word and sentence representations and feeding them into CNN layers for higher-level feature extraction. Their hybrid pipeline demonstrated how PLM embeddings can be stacked with hierarchical convolutional reasoning to exploit multiple levels simultaneously. Bama et al.[45] integrated word and sentence-level embeddings in a hierarchical transformer with label attention mechanisms. Their model selectively aligned features to specific trait labels, effectively linking multi-level linguistic signals to psychological categories for MBTI classification. Ramezani et al. [24] advanced a more comprehensive framework that explicitly integrated lexical, semantic, and sequential levels. Their ensemble combined term-frequency vectorization (lexical), ontology-based and enriched ontology models (semantic), latent semantic analysis (document-level semantic abstraction), and BiLSTM-based deep learning (sequential word-level). These base models were stacked into a meta-model using a Hierarchical Attention Network, which performed reasoning across levels. This ensemble highlights how multiple modeling strategies can be fused into a unified multi-level pipeline. Yang et al.[46] addressed sentence and document-level interactions in their DeepPerson model. They integrated hierarchical attention networks with pre-trained language models, allowing sentence-level interactions to inform document-level personality representations. Importantly, their framework incorporated psychological theory, positioning multi-level modeling not only as a computational choice but also as a way to align with established psychological constructs. Yang et al.[47] proposed Transformer-MD, which combined sentence and document-level modeling. By employing Transformer-XL memory tokens, their approach captured dependencies across multiple short user-generated posts at the document level while retaining sentence-level contextual modeling within individual posts. This dual integration addressed the fragmented nature of social media text.

Liu et al. [48] developed a compositional and language independent model for personality trait recognition that combined word and sentence-level features. They encoded tweets in multiple languages using distributed embeddings and aggregated them via a hierarchical composition function to predict Big Five traits. By leveraging both local lexical cues and sentence-level structure, the model could generalize across languages and short-text social media settings, highlighting the value of multi-level representations in low-resource scenarios.

Lynn et al. [49], proposed a hierarchical approach for user personality prediction by modeling word, message, and document-level representations. They employed attention mechanisms at the message level to capture the relative importance of individual posts within a user's social media history. This hierarchical structure allowed the model to integrate fine-grained lexical signals with broader user-level behavioral patterns, improving Big Five trait inference compared to flat, single-level baselines.

Tirotta et al. [50] introduced a multi-level framework that integrates sentence and document-level embeddings for personality prediction. They extracted sentence embeddings from short social media posts using contextual language models and then aggregated these into document-level representations to infer Big Five traits. This combination allowed the model to capture both fine-grained linguistic patterns and cross-sentence dependencies, demonstrating that sentence-level embeddings can be effectively composed to enhance document-level personality inference.

Taken together, these studies show that multi-level approaches systematically bridge signals from words, sentences, and documents rather than collapsing text into a flat representation. Their strength lies in capturing hierarchical dependencies that reflect how language is naturally structured, often enriched by psycholinguistic knowledge or external ontologies. This results in improved contextual awareness and interpretability compared to single-level methods. However, these models also come with notable challenges: increased computational complexity, reliance on external resources (such as lexicons or ontologies), and difficulty in balancing contributions from different levels without introducing redundancy.

Multi-level text-based personality prediction represents a methodological advance over single-level approaches by explicitly modeling hierarchical linguistic structure. By linking fine-grained cues at each context level, these methods more faithfully capture the layered nature of personality expression in language. At the same time, they highlight the open research challenge of designing efficient architectures that can exploit multi-level signals without sacrificing scalability or overfitting.

2.3 Research Gaps

Despite the range of methods reviewed in the preceding categories (single-level approaches 2.1 and multi-level approaches 2.2), several critical gaps remain. First, although some studies are categorized as multi-level, no existing framework fully exploits the hierarchical nature of text across all syntactic levels (namely word, sentence, and document) within a unified model. Second, hypergraph representations, which naturally capture multi-level dependencies by extending beyond pairwise relations and encoding hierarchical structures [51], have received little to no attention in Automatic Personality Prediction (APP) [52]. Third, only a limited number of studies have achieved satisfactory results relying solely on the Essays dataset, which remains the most widely used benchmark in personality prediction research.

This work addresses these gaps through a novel multi-level pipeline. We represent text as a hypergraph spanning document, sentence, and word levels, then transform it into a hierarchical graph to bridge hypergraph structures with graph neural network (GNN) processing. A learnable feature-selection module, combined with a transformer-based graph encoder, enables joint representation learning and dimensionality reduction. Importantly, our method relies exclusively on the Essays dataset and achieves improved predictive performance through a hierarchical, multi-level learning strategy.

Name	Integrated Level	Models	Dataset	Personality Model	Context	Year
Ong et al. [29]	document	Support Vector Machine (SVM), XGBoost	Twitter	Big Five	Social media text	2017
Shen et al. [30]	document	Joint, sequential, survival generative models	Enron Email Corpus	Big Five	Email	2013
Skowron et al. [31]	document	Multimodal Personality Regressors	Socila media	Big Five	Social media profiles	2016
Jayaraman et al. [32]	document	XLNet	Essays dataset	Big Five	Essays	2024
Christian et al. [19]	document	Model averaging (BERT, RoBERTa, and XLNet)	Social media	Big Five	Social media posts	2021
Hao Lin and Xiaolei Li [33]	document	Pre-trained Language Models, SenticNet6	Weibo	Big Five	Social media text	2023
Kamalesh and Bharathi [34]	document	BERT, LSTM	Social media	Big Five	Facebook status	2022
V Ganesan et al. [35]	doucment	GPT-3, WT-LEX, MFC (Most Frequent Class)	Social media	Big Five	Socila media posts	2023
Naz et al. [36]	sentence	Bi-LSTM	Essays dataset	Openness trait	Essays	2025
Leonardi et al. [37]	sentence	Transformer-based deep neural network	Social media	Big Five	Facebook posts	2020
Kazameini et al. [38]	word	Bagged SVM, BERT	Essays dataset	Big Five	Social media and essay text	2020
Alsini et al. [39]	word	Transformer based models, Bi-LSTM	MBTI dataset	Agreeableness trait	MBTI dataset text	2024

Table 1: Single-level text-based approaches

Name	Integrated Levels	Models	Dataset	Personality Model	Context	Year
Zhu et al. [25]	word, document	Graph Convolutional Networks (GCNs)	MyPersonality, YouTube, PAN2015	MBTI	User data	2024
Zhu et al. [41]	word, document	Heterogeneous Message-Passing Graph Neural Network (HPMN),	Essays dataset, psycholinguistic lexicon	Big Five	lexicon-guided Essays dataset text	2022
Johnson and Murty [42]	word, document	CNN, RNN, LSTM, BiLSTM	Social media, DBpedia, ontology	Big Five	Social media text	2023
Kerz et al. [43]	word, document	BERT, BiLSTM	Essays dataset	Big Five	Essays dataset text	2023
Salahat et al. [44]	word, sentence	Convolutional Neural Networks (CNNs)	Social media	Big Five	Social media posts	2023
Bama et al. [45]	word, sentence	Hierarchical Transformer network with Label Attention for Personality Prediction (HT-LA-PP)	MBTI dataset (social media)	MBTI	Social media posts	2025
Ramezani et al. [24]	word, document	Hierarchical Attention Network (HAN)	Essays dataset, ontology	Big Five	Essays dataset text	2022
Yang et al. [46]	sentence, document	Hierarchical Attention Network (HAN)	Essays dataset	Big Five	Essays dataset text	2023
Yang et al. [47]	sentence, document	Transformer-MD, Dimension Attention	Social media	Big Five	Social media posts	2021
Liu et al. [48]	word, sentence	Bi-GRU	Twitter	Big Five	Tweets in multiple languages	2016
Lynn et al. [49]	word, sentence, document	Hierarchical Attention Network (HAN)	Social media	Big Five	Social media posts (multiple per user)	2020
Tirotta et al. [50]	sentence, document	SBERT	Twitter, MNLI (as auxiliary)	Big Five	Short posts and tweets	2023

Table 2: Multi-level text-based approaches

3 Methodology

3.1 Dataset

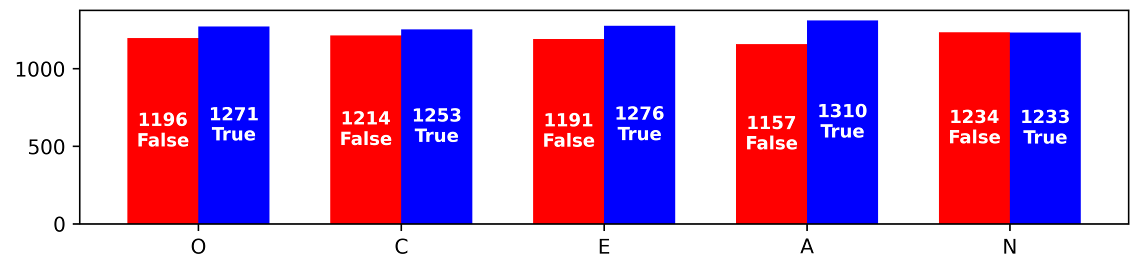
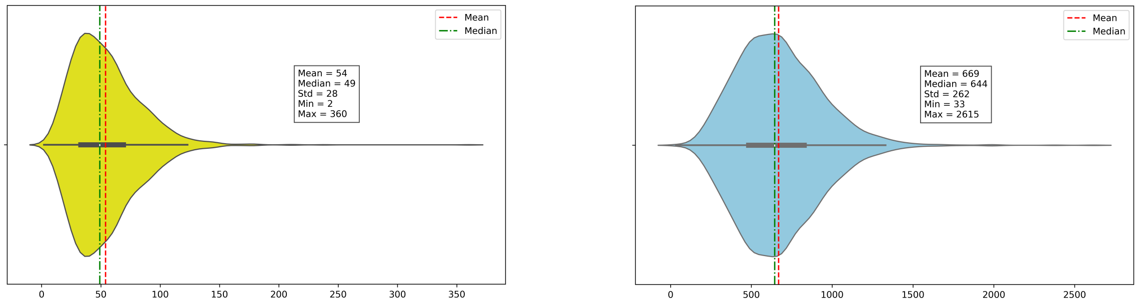


Figure 1: Label Distribution for Big Five Personality Traits in the Essays Dataset

The textual data used in this work was extracted from the *Essays* dataset, introduced by Pennebaker, and King [53], which comprises over self-authored essays, labeled with Big-Five personality scores from self-assessments. Due to its free-form structure, richness of expression, and direct personality annotations, it has become a benchmark for evaluating text-based models of Automatic Personality Prediction (APP). The dataset was developed as part of a psychological research program aimed at examining whether language use reflects stable personality traits. The dataset is grounded in a word-based, computerized text analysis approach and was among the first large empirical efforts to link natural language to psychological characteristics, particularly those defined by the Big Five personality traits.

The Essays dataset has since become a widely used benchmark in fields such as personality prediction, natural language processing, and computational psycholinguistics. It contains a total of 2468 short essays, written by over 1,200 undergraduate students, who each completed four writing assignments (two at the beginning and two at the end of a semester). They were instructed to write on a variety of unrestricted topics such as stream of consciousness writing, and reflections on coming to college. Each essay was approximately 500-800 words, with no strict length, grammar, or stylistic constraints (students were simply asked to write freely).

The bar chart in Figure 1, illustrates the distribution of binary personality labels in the Essays dataset, where each trait in the Big Five model (Openness (O), Conscientiousness (C), Extraversion (E), Agreeableness (A), and Neuroticism (N)) is labeled as either True (0) or False (1). The distribution is relatively balanced across traits, which is ideal for multi-label classification tasks. For Openness, there are 1,196 samples labeled as false and 1,271 as true, indicating a slightly higher presence of open-minded individuals. Conscientiousness has 1,214 false and 1,253 true samples, also showing a mild skew toward presence. Extraversion is labeled true in 1,276 and false in 1,191 cases, suggesting a majority of extroverted participants. Agreeableness exhibits the largest gap, with 1,310 present and 1,157 false



A Sentence Distribution

B Word Distribution

Figure 2: Sentence(A) and Word(B) Distribution in the Essays Dataset

labels, pointing to a generally cooperative and trusting population. Neuroticism is almost perfectly balanced with 1,234 false and 1,233 true cases, offering a symmetric distribution for this trait. This overall balance across traits helps reduce bias and supports reliable training and evaluation of personality prediction models.

Figure 2 depicts the distributions of the documents in terms of number of words and number of sentences, respectively. The sentence count distribution (Figure 2B) has a right-skewed distribution, with a mean of 54 and a median of 49 sentences per document and the standard deviation (Std) is 28, indicating moderate variation in sentence count. The minimum sentence count is 2 and the maximum is 360, showing the existence of outlier documents with extremely high sentence counts. The distribution's bulk lies within the 30-70 sentence range, and the slight skew is supported by the mean being higher than the median.

The distribution of word counts(Figure 2B), also shows a moderately right-skewed distribution, with a mean of 669 and a median of 644 words per document and the standard deviation (Std) is 262, indicating considerable variability in document length. The minimum word count is 33, and the maximum reaches 2,615 words, highlighting the presence of a few significantly longer documents that contribute to the skewness. The relatively small difference between the mean and median suggests the presence of mild outliers rather than extreme ones. The density is concentrated around the 500-800 range, where most documents fall.

The statistics in Figure 2 indicate that while most documents are relatively similar in length, there is a long tail in both distributions which needs sophisticated methods to adapt towards downstream tasks.

3.2 Proposed Method

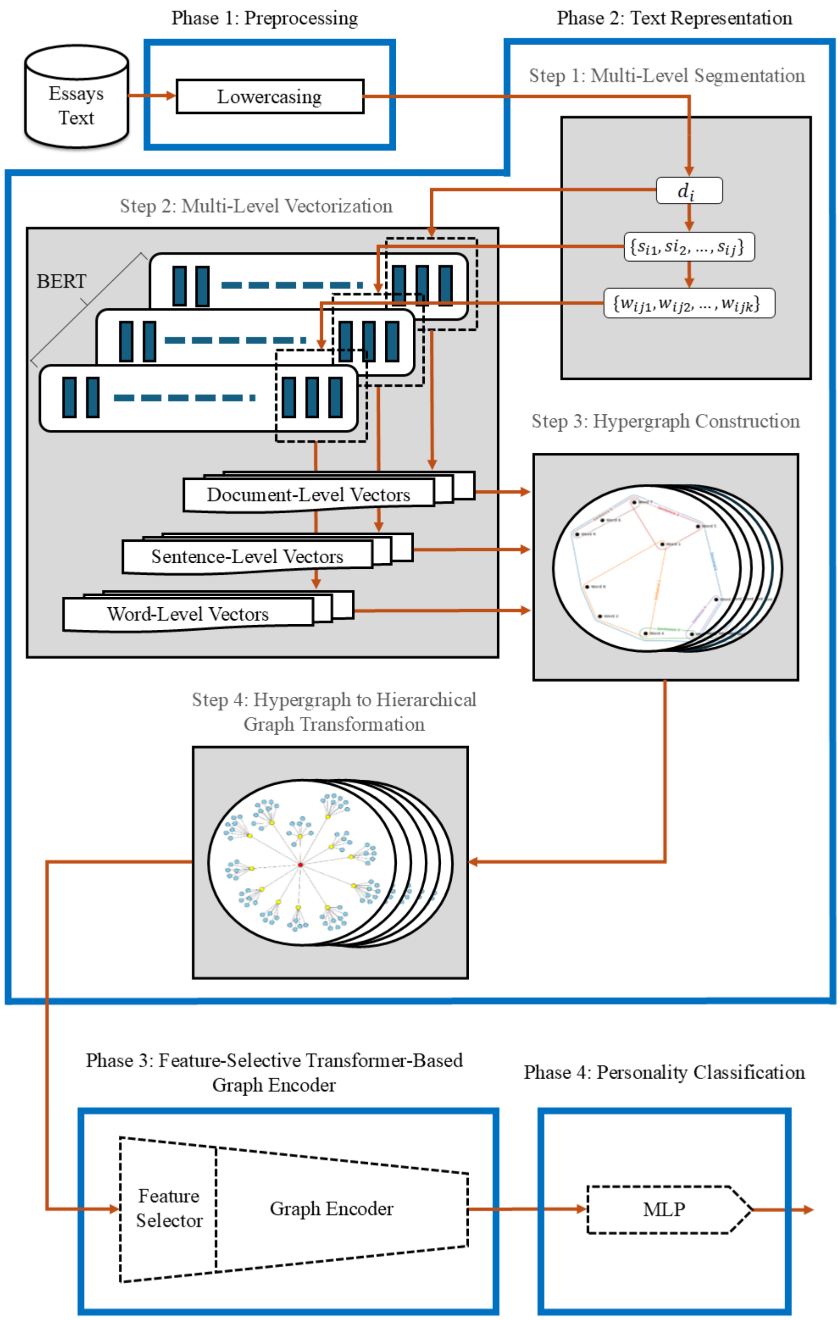


Figure 3: The overall structure of Hyper Persona

We propose **Hyper Persona**, a **transformer-based graph encoder** designed to **predict personality** from **text-driven multi-level hypergraphs**. The pipeline begins by preprocessing the textual data, which is then segmented according to the following hierarchy: *document-level, sentence-level, and word-level.* Each segment is vectorized to obtain mathematical representations at these three syntactic levels. Using these vectors, we construct multi-level hypergraphs that capture the hierarchical structure of the text. These hypergraphs are subsequently transformed into hierarchical graphs and labeled with the corresponding personality traits. The graphs are processed by the transformer-based graph encoder for effective representation learning and dimensionality reduction. A fixed-size graph-level representation is then generated in a lower-dimensional space. Finally, a lightweight multi-layer perceptron (MLP) classifier is trained to predict personality traits based on the Big Five model.

As illustrated in Figure 3, our pipeline consists of four main phases: Phase 1: Preprocessing (3.2.1), Phase 2: Text Representation (3.2.2), Phase 3: Representation Learning and Dimensionality Reduction (3.2.3), and Phase 4: Personality Classification (3.2.4).

3.2.1 Phase 1: Preprocessing

We applied minimal preprocessing to preserve the linguistic richness. Specifically, we performed lower casing to ensure consistency, while deliberately avoiding other common preprocessing techniques. This decision aligns with the transformer-based nature of our *multi-level vectorization 3.2.2* method, which has enabled us to work with raw text and benefit from the retention of both syntactic and semantic information.

3.2.2 Phase 2: Text Representation

The phase 2 of our pipeline is called *text representation*, including four core steps, step 1: multi-level segmentation (3.2.2.1), step 2: multi-level vectorization (3.2.2.2), step 3: hypergraph construction (3.2.2.3), step 4: hypergraph to hierarchical graph transformation (3.2.2.4), which will be explained in detal.

3.2.2.1 *Step 1: Multi-level Segmentation*

As detailed in Algorithm 1, the *multi-level segmentation* procedure begins by taking the corpus $\mathcal{C}$ as input and initializing an empty list $\mathcal{SC}$ to store the segmented output. The algorithm iterates over each document $d_i \in \mathcal{C}$. Eeach document is stored in $\mathcal{D}_i$, which stands for document-level segment, the sentence-level segmentation is performed using the function $segment\_to\_sentences(d_i)$, producing a list of sentences $\mathcal{S}_{d_i}$. For each sentence $s_j \in \mathcal{S}_{d_i}$, word-level segmentation is performed via $segment\_to\_words(s_j)$, resulting in word sequences that are accumulated into a list $\mathcal{W}s_j d_i$, which preserves the grouping of words in the sentence s_j.

Algorithm 1 Multi-level Segmentation

function SEGMENT($\mathcal{C}$) ▷ Take $\mathcal{C}$ ($\mathcal{C}$orpus) as input
    $\mathcal{SC} \leftarrow [\,]$ ▷ Initialize the result list as $\mathcal{SC}$ ($\mathcal{S}$egmented $\mathcal{C}$orpus)
    for all $d_i \in \mathcal{C}$ **do** ▷ Iterate over each document
        $\mathcal{D}_i \leftarrow d_i$
        $\mathcal{S}_{d_i} \leftarrow \textit{segment\_to\_sentences}(d_i)$ ▷ Segment document into sentences
        $\mathcal{W}_{s_j d_i} \leftarrow [\,]$
        for all $s_j \in \mathcal{S}_{d_i}$ **do**
            Add $\textit{segment\_to\_words}(s_j)$ to $\mathcal{W}_{s_j d_i}$ ▷ Segment sentece into words
        end for
        $\mathcal{SD}_i \leftarrow (\mathcal{D}_i, (\mathcal{S}_{d_i}, (\mathcal{W}_{s_j d_i})))$
        Add $\mathcal{SD}_i$ to $\mathcal{SC}$
    end for
    return $\mathcal{SC}$ ▷ Return segmented corpus
end function

After both multi-level segmentation, the algorithm constructs a nested hierarchical structure $(\mathcal{D}i, (\mathcal{S}_{d_i}, (\mathcal{W}s_j d_i)))$, and sotres it in $\mathcal{SD}_i$, which encapsulates all three syntactic levels (document, sentence, and word), and is added to the output list $\mathcal{SC}$. Upon completion, the algorithm returns $\mathcal{SC}$, a hierarchically segmented version of the input corpus. Each element in $\mathcal{SC}$ is a segmented representation of a document in our original copus $\mathcal{C}$.

3.2.2.2 *Step 2: Multi-level Vectorization*

To encode textual data at different levels, we designed a **multi-level vectorization** strategy to embed text at multi-syntactic levels we made in *multi-level segmentation 3.2.2.1*. Specifically, we applied *layer aggregation* strategy to the *BERT* pretrained model [14], which has enabled us to extract contextual and positional representations from the model in different syntactic-levels, using the segments we made in *multi-level segmentation* (3.2.2.1).

layer aggregation: In transformer-based language models, each input token is represented by a hidden state vector at every layer of the model. These hidden states capture different levels of linguistic and semantic information as the input passes through the layers. The upper layers of these embeddings produced more context-specific representations than the lower layers. Less than 5% of the variance on average, in contextualized representations, might be defined by static embedding [54].

Our multi-level vectorization technique relies on a layer averaging strategy, where we average the hidden states of several upper transformer layers of BERT. Formally, for input text $\mathcal{T}$, we define the embedding function as:

Algorithm 2 Multi-level Vectorization

function VECTORIZE($\mathcal{SC}$) ▷ Take $\mathcal{SC}$ ($\mathcal{S}$egmented $\mathcal{C}$orpus) as input
  $\mathcal{VC} \leftarrow [\ ]$
  for all $\mathcal{SD}_i \in \mathcal{SC}$ **do**
    $\mathcal{VD}_i \leftarrow Embed(\mathcal{D}_i)$
    $\mathcal{SV}_{d_i} \leftarrow [\ ]$
    for all $s_j \in \mathcal{S}_{d_i}$ **do**
      $vs_j \leftarrow Embed(s_j)$
      Add vs_j to $\mathcal{SV}_{d_i}$
      $\mathcal{WV}_{s_j d_i} \leftarrow [\ ]$
      for all $w_k \in \mathcal{W}_{s_j d_i}$ **do**
        $vw_k \leftarrow Embed(w_k)$
        Add vw_k to $\mathcal{WV}_{s_j d_i}$
      end for
    end for
    Add $(\mathcal{VD}_i, (\mathcal{SV}_{d_i}, (\mathcal{WV}_{s_j d_i})))$ to VC
  end for
  return $\mathcal{VC}$
end function

$$\text{Embed}(\mathcal{T}) = \frac{1}{L} \sum_{l=l_1}^{l_2} \mathbf{h}^{(l)}(\mathcal{T}), \tag{1}$$

where $\mathbf{h}^{(l)}(\mathcal{T})$ is the hidden state from layer (l) for the input text $\mathcal{T}$, and l_1, l_2 are the lowest and highest target layers of the model, which makes L the number of layers aggregated respectively. Aggregating hidden states across multiple transformer layers is a technique to enhance the quality of contextual embeddings. This approach allows the model to fuse information from different depths, where lower layers typically capture syntactic and lexical features, and higher layers encode more abstract and semantic patterns. By averaging the hidden states over a range of layers, the resulting representation benefits from richer linguistic features and reduced sensitivity to noise from any single layer.

Empirically, it has been shown that aggregating the top layers of transformer models often improves performance on downstream tasks such as text classification, semantic similarity, and representation learning. This layer aggregation strategy produces a comprehensive and semantically meaningful embedding for the input text, which is the core of our multi-level vectorization technique.

Algorithm 2 presents the Multi-level Vectorization procedure, transforming our segmented corpus $\mathcal{SC}$ into a hierarchical vectorized structure. The input $\mathcal{SC}$ is our hierarchically segmented corpus as previously described in the *multi-level segmentation 3.2.2.1*. For each

Algorithm 3 Hypergraph Construction

function ConstructHypergraphs($\mathcal{VC}$) ▷ $\mathcal{VC}$ ($\mathcal{V}$ectorized $\mathcal{C}$orpus)
    $\mathcal{HG} \leftarrow [\,]$
    for each $\mathcal{VD}_i$ in $\mathcal{VC}$ **do**
        Initialize $\mathcal{HG}_i$ ▷ $\mathcal{HG}_i$ (hypergraph object for document i)
        $hypergraph_i \leftarrow v_{d_i}$
        Add $hypergraph_i$ to $\mathcal{HG}_i$
        for each sv_j in $\mathcal{SV}_{d_i}$ **do**
            $hyperedge_{s_j} \leftarrow sv_j$
            Add $hyperedge_{s_j}$ to $\mathcal{HG}_i$
            for each wv_k in $\mathcal{WV}_{s_j d_i}$ **do**
                $node_k \leftarrow wv_k$
                Add $node_k$ to $\mathcal{HG}_i$
            end for
        end for
        Add $\mathcal{HG}_i$ to $\mathcal{HG}$
    end for
    return $\mathcal{HG}$
end function

segmented document $\mathcal{SD}i \in \mathcal{SC}$, a document-level embedding $\mathcal{VD}_i$ is computed using the $Embed()$ function in Eq 1. Subsequently, each sentence s_j in the document is individually embedded to produce a sentence-level embedding vs_j, which is collected into $\mathcal{SV}d_i$. For each sentence s_j, its constituent words w_k are also embedded to obtain word-level vectors vw_k, which are collected into $\mathcal{WV}s_jd_i$.

The output is a three-level nested structure $\mathcal{VC}$, where each entry consists of a document vector $\mathcal{VD}_i$, its corresponding sentence vectors $\mathcal{SV}_{d_i}$, and for each sentence, the associated word vectors $\mathcal{WV}_{s_jd_i}$. This hierarchical representation captures the document's structure and semantics across three granularities, enabling us to leverage multi-level textual features.

3.2.2.3 *Step 3: Hypergraph Construction*

To capture the hierarchical structure of text, we construct a multi-level hypergraph for each document. In this representation, we treat both the document and each sentence as hyperedges, while words serve as the fundamental nodes. Specifically, each sentence hyperedge encompasses its constituent word nodes, and the document hyperedge embraces all its sentence-level embeddings. Each node (word) and each hyperedge (sentences and document) is associated with a contextual embedding vector derived from BERT mentioned in step 2 (3.2.2.2).

Algorithm 3 outlines the process for constructing a multi-level hypergraph from hierarchical text vectors in step 2 (3.2.2.2). The input to the algorithm is $\mathcal{VC}$, which contains the document embedding $\mathbf{E}_{\mathcal{D}_i}$ and a list of sentence-level information, where each item is a pair of a sentence embedding $\mathbf{E}_{S_{ij}}$ and a list of word embeddings $\mathbf{E}_{\mathcal{W}_{ijk}}$. First, a hyperedge representing the document ($hyperedge_{D_i}$) is created, and its embedding is stored as a feature. The algorithm then iterates over each sentence, adding a new sentence-level hyperedge ($hyperedge_{S_{ij}}$) with its corresponding embedding vector as feature. For every word within a sentence, a node ($node_{W_{ijk}}$) is added to the hypergraph with its own embedding vector as feature. This algorithm results in a structured hypergraph shown in Figure 4A, that captures both intra-level (sentence to words) and inter-level (document to sentences) hierarchy, enabling expressive modeling of semantic relationships across textual levels.

3.2.2.4 *Step 4: Hypergraph to Hierarchical Graph Transformation*

To bridge the gap between hypergraph structures and graph neural networks, we convert our hypergraphs (Figure 4A) into hierarchical graphs (Figure 4B).

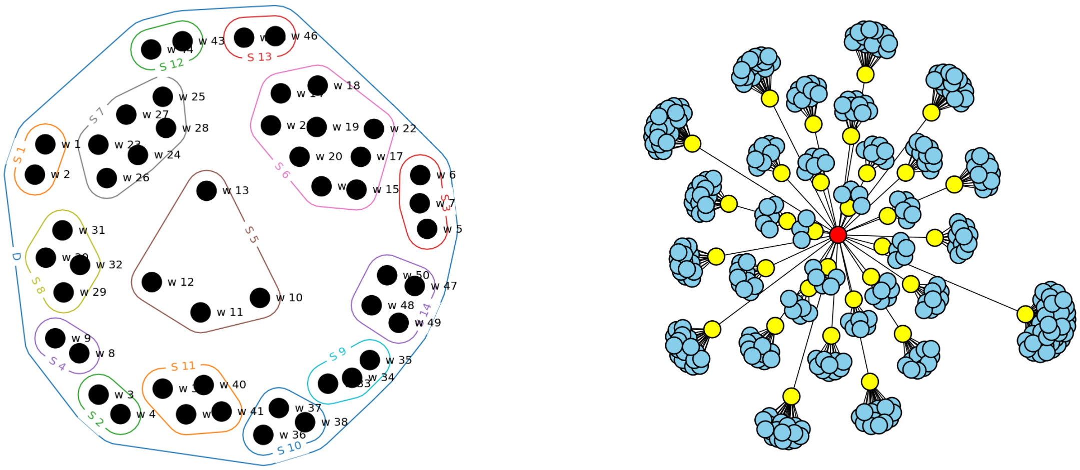
In the original hypergraph (Figure 4A), nodes represent words w_j, and hyperedges represent sentences S_i, surrounded with a higher-order hyperedge representing the entire document D. Each sentence hyperedge connects a subset of words, and the document hyperedge connects all sentences in the document. To overcome the constraint of pair wise relations, which is the default assumption of graph neural networks, while preserving the hypergraph's semantic hierarchy, we transform each hypergraph to a hierarchical graph structure (Figure 4B).

Firstly, the document node (red node in 4B) is placed in the core representing the document hyperedge and is eqquipped with its corresponding embedding vector as node feature. Then the sentence nodes (yellow nodes in 4B) are attached to the document nodes with their corresponding embedding vectors as node feature. The word nodes (blue nodes in 4B) are placed and attached to their parent sentences afterwards, and the corresponding embedding vectors are added as their node feature. The undirected edges connecting our nodes in the hierarchical graphs, are eqquipped with similarity scores, which are calculated through cosine similarity, between two nodes that connect through an edge.

3.2.3 Phase 3: Feature-Selective Transformer-Based Graph Encoder

3.2.3.1 Feature Selection

In the *feature-selective transformer-based graph encoder*, we introduce a *learnable feature selection* mechanism at the input stage to identify the most informative node attributes for downstream personality prediction. A learnable weight vector $\mathbf{s} \in \mathbb{R}^d$, where d is the number of node features, generates a differentiable hard feature mask using the *gumbel-sigmoid* strategy. The resulting binary mask stochastically selects a subset of features, effectively suppressing irrelevant dimensions while remaining fully differentiable:



A Hypergraph B Hierarchical Graph

Figure 4: Illustration of a Hypergraph (A), and a Hierarchical Graph (B)

$$\mathcal{M} = \text{GumbleSigmoid}(\mathbf{s}/\tau), \tag{2}$$

where $\mathcal{M}$ is the feature mask, and τ is the temperature parameter controlling the sharpness of the selection. The masked input is obtained as:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i \odot \mathbf{m}, \tag{3}$$

where $\odot$ denotes element-wise multiplication. This adaptive feature selection reduces noise and promotes sparsity in the learned representations.

3.2.3.2 Gumbel-Sigmoid

The *Gumbel-Sigmoid* is a continuous relaxation of discrete *Bernoulli sampling*, enabling stochastic binary decisions to remain differentiable and thus trainable with backpropagation. It leverages the *Gumbel-Max trick*, which samples from a categorical distribution by perturbing logits with Gumbel noise and selecting the maximum. For the binary case, this mechanism is adapted into a sigmoid form. Concretely, for each feature weight s_j in the learnable vector $\mathbf{s}$, the Gumbel-Sigmoid generates a sample according to:

$$m_j = \sigma\left(\frac{s_j + g_j}{\tau}\right), \quad g_j \sim \text{Gumbel}(0, 1), \tag{4}$$

where g_j is a random variable sampled from the standard Gumbel distribution, τ is the temperature parameter, and $\sigma(\cdot)$ denotes the logistic sigmoid function. The temperature τ

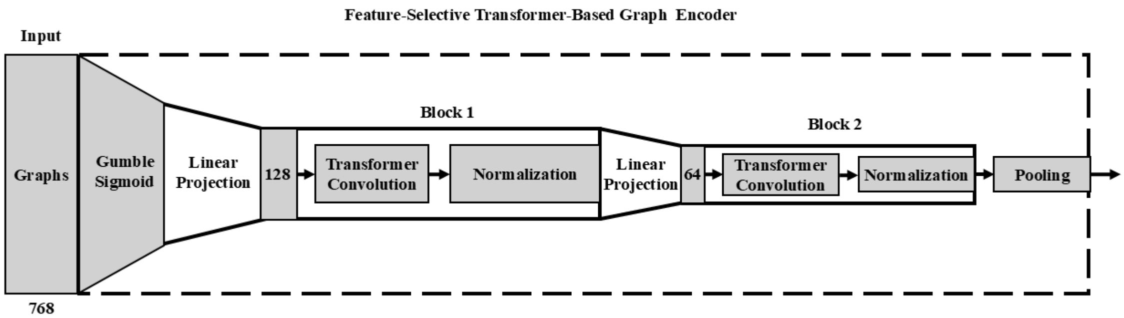


Figure 5: Architecture of the Feature-Selective Transformer-Based Graph Encoder

controls the smoothness of the approximation: as $\tau \to 0$, the distribution becomes nearly discrete, producing binary-like selections that approximate hard masking, whereas higher τ values yield softer, probabilistic masks.

This property allows the model to transition from *exploration* (soft, noisy selections at higher τ) to *exploitation* (sharp, deterministic selections at lower τ) during training. Importantly, the Gumbel-Sigmoid trick preserves differentiability despite producing near-binary masks. Gradients flow through the continuous relaxation, allowing the feature selection mask $\mathcal{M}$ to be optimized jointly with the graph encoder via standard backpropagation. This enables end-to-end learning of which input features are most relevant while suppressing noisy or redundant dimensions without requiring explicit non-differentiable thresholding.

3.2.3.3 Transformer-Based Graph Encoder

The backbone of the encoder is a two-block transformer-based graph encoder that performs attention-driven message passing over graph nodes. Unlike conventional graph convolutional operators, which aggregate neighbor features with fixed or degree-normalized weights, the TransformerConv operator leverages *self-attention* to adaptively weight neighbor contributions based on their contextual relevance. This design enables the encoder to integrate both *local structural dependencies* and *semantic correlations* among connected nodes.

At layer l, the hidden representation of node i is updated as:

$$\mathbf{h}_i^{(l)} = \mathrm{Sigmoid}\left(\mathrm{LayerNorm}\Big(\mathbf{W}_{\mathrm{res}}^{(l)}\mathbf{h}_i^{(l-1)} + \mathrm{TransformerConv}^{(l)}(\mathbf{h}_i^{(l-1)})\Big)\right), \tag{5}$$

where $\mathbf{h}_i^{(0)} = \tilde{\mathbf{x}}_i$ corresponds to the masked input after feature selection, and $\mathbf{W}_{\mathrm{res}}^{(l)}$ is a residual projection matrix that aligns dimensions between successive layers, ensuring stable gradient flow. The use of *residual connections* mitigates vanishing gradients and allows the model to preserve essential node-level signals across deeper layers. The *LayerNorm* term standardizes feature statistics within each node embedding, thereby reducing covariate shift and accelerating training convergence. Finally, a sigmoid nonlinearity is applied to bound the latent representations into a normalized range, which helps stabilize subsequent pooling and classification.

The core operator, TransformerConv$^{(l)}$, extends the message-passing paradigm with a scaled dot-product attention mechanism. For node i, the update rule is expressed as:

$$\mathbf{x}'_i = \mathbf{W}_1 \mathbf{x}_i + \sum_{j \in \mathcal{N}(i)} \alpha_{i,j} \mathbf{W}_2 \mathbf{x}_j, \tag{6}$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i. The first term $\mathbf{W}_1 \mathbf{x}_i$ preserves self-information through a learnable transformation, while the second term aggregates neighbor information, modulated by the attention coefficient $\alpha_{i,j}$. This coefficient reflects the importance of neighbor j in updating node i, and is computed via multi-head scaled dot-product attention:

$$\alpha_{i,j} = \text{softmax}\left(\frac{(\mathbf{W}_3 \mathbf{x}_i)^\top (\mathbf{W}_4 \mathbf{x}_j)}{\sqrt{d}} \right). \tag{7}$$

Here, $\mathbf{W}_3$ and $\mathbf{W}_4$ project node features into query and key spaces, respectively, while the denominator $\sqrt{d}$ normalizes the inner product to avoid sharp gradients in high-dimensional settings. The softmax function ensures that the attention coefficients form a probability distribution over the neighbors of i, thereby enforcing relative weighting of structural dependencies. Multi-head attention can be incorporated by repeating this computation across several independent subspaces and either concatenating or averaging the results, though in our implementation we use a single head to maintain computational efficiency and interpretability.

Intuitively, this mechanism allows each node to selectively focus on the most semantically aligned neighbors, rather than treating all neighbors as equally informative. Unlike traditional GCN-style averaging, the TransformerConv operator dynamically adapts its receptive field based on both node content and graph structure. This is particularly important in text-derived graphs, where syntactic neighbors may carry varying degrees of importance depending on context. As a result, the encoder is able to integrate hierarchical linguistic cues while simultaneously suppressing noisy or weakly correlated signals.

Graph-Level Representation: After two transformer blocks, the node embeddings capture both local structural dependencies and attention-weighted contextual signals. To obtain a holistic representation of the entire graph, we employ a *global additive pooling* operation that aggregates the final node embeddings into a fixed-dimensional vector:

$$\mathbf{z}_G = \sum_{i \in G} \mathbf{h}_i^{(2)}, \tag{8}$$

where $\mathbf{h}_i^{(2)}$ denotes the final embedding of node i after the second TransformerConv block. This summation is permutation-invariant, meaning that the representation is unaffected by the ordering of nodes in the input graph, a property essential for processing arbitrary text-derived graphs where node order has no intrinsic meaning.

The choice of additive pooling is motivated by its ability to preserve cumulative contributions from all nodes, thereby capturing the overall distribution of linguistic and structural cues present in the graph. Unlike max pooling, which retains only the most dominant feature per dimension, or mean pooling, which normalizes contributions and may dilute strong signals, summation pooling allows highly informative nodes to exert proportionally greater influence on the graph representation. This is particularly beneficial in the context of personality prediction, where certain nodes (words or sentences with strong psychological markers) should contribute more prominently to the final representation.

Formally, additive pooling can be interpreted as learning a bag-of-embeddings representation of the graph, where the magnitude of $\mathbf{z}_G$ reflects both the strength and frequency of node-level signals. This aggregated vector thus encodes both feature information (via learned node embeddings) and structural information (via attention-driven message passing), making it well-suited for downstream tasks. The resulting graph-level embedding $\mathbf{z}_G$ serves as the input to the personality classification head, where it is mapped to predictions of the Big-Five trait scores.

3.2.4 Phase 4: Personality Classifier

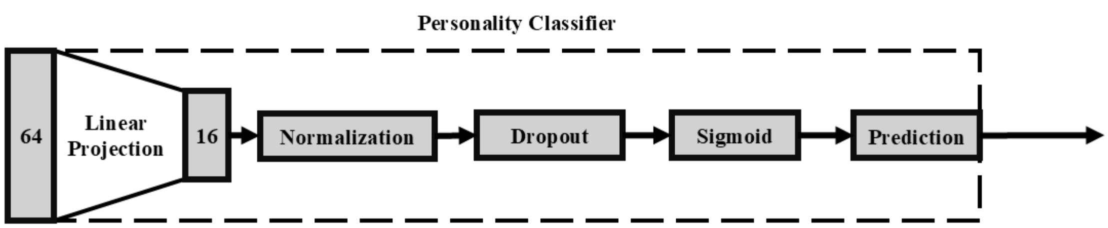


Figure 6: Architecture of the Personality Classifier with Sigmoid-Gated Latent Representation

The personality classifier maps the pooled graph embedding $\mathbf{z}_G \in \mathbb{R}^{64}$ into a compact latent space before producing label prediction for the Big Five trait. This classification head is designed to be lightweight yet expressive, ensuring that the personality prediction is driven primarily by the structural and semantic representations learned in the graph encoder while still benefiting from regularization and non-linear transformations.

The forward pass through the classifier proceeds as follows:

$$\mathbf{z}_1 = \mathbf{W}_1 \mathbf{z}_G + \mathbf{b}_1, \quad \mathbf{z}_1 \in \mathbb{R}^{16}, \tag{9}$$

$$\mathbf{z}_2 = \mathrm{LayerNorm}(\mathbf{z}_1), \tag{10}$$

$$\mathbf{z}_3 = \mathrm{Dropout}(\mathbf{z}_2), \tag{11}$$

$$\mathbf{z}_4 = \sigma(\mathbf{z}_3), \tag{12}$$

$$\hat{\mathbf{y}} = \mathbf{W}_2 \mathbf{z}_4 + \mathbf{b}_2, \quad \hat{\mathbf{y}} \in \mathbb{R}^1, \tag{13}$$

where $\mathbf{W}_1, \mathbf{W}_2$ and $\mathbf{b}_1, \mathbf{b}_2$ are learnable projection parameters. The intermediate latent activation $\mathbf{z}_4$ acts as a sigmoid-gated bottleneck representation, while $\hat{\mathbf{y}}$ corresponds to the raw logits for the five personality traits.

Layer normalization stabilizes hidden activations and prevents internal covariate shift, which is crucial in small latent spaces where fluctuations in scale can disproportionately affect prediction quality. The subsequent *dropout* layer provides regularization by randomly zeroing elements of the normalized vector, reducing overfitting and encouraging the model to rely on distributed representations rather than memorizing spurious correlations. The inclusion of a *sigmoid nonlinearity* ensures that the latent features are bounded between $[0, 1]$, which can be interpreted as gating the latent representation before projection to the trait space. This design adds an additional layer of non-linearity, improving the expressive power of the classifier without adding significant computational overhead.

The final linear projection maps the gated latent representation into the output space of the label. Since personality prediction is inherently a *binary classification problem*, the network is optimized using the binary cross-entropy (BCE) loss:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{5} \sum_{k=1}^{5} \Big(y_k \log \sigma(\hat{y}_k) + (1 - y_k) \log(1 - \sigma(\hat{y}_k)) \Big), \tag{14}$$

where $y_k \in \{0, 1\}$ is the ground truth label for trait k, and $\sigma(\cdot)$ is the logistic sigmoid function applied only during loss computation. This ensures numerical stability and allows the network to jointly optimize all five traits in a unified framework.

By combining compact linear projections, normalization, dropout regularization, and a sigmoid-gated latent representation, the classifier effectively bridges the high-dimensional graph embeddings with the interpretable predictions. This modular design ensures that the bulk of representational learning is handled by the graph encoder, while the classifier enforces robustness and facilitates efficient binary label mapping.

4 Results & Discussion

4.1 Evaluation Metrics

To comprehensively assess the performance of the binary classification model, four complementary evaluation metrics are employed: Accuracy 4.1.1, Precision 4.1.3, Recall 4.1.4, and the F1-score 4.1.2. These measures collectively capture the global correctness of predictions, providing a robust characterization of model behavior under potential class imbalance. All metrics are computed based on the confusion matrix components (true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN)).

4.1.1 Accuracy

Accuracy quantifies the overall proportion of correctly classified instances, regardless of class.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}. \tag{15}$$

This measure provides an intuitive summary of performance by indicating the fraction of correct predictions across all samples. However, in imbalanced datasets where one class is significantly more frequent than the other, accuracy alone can be misleading, as a model may achieve high accuracy simply by favoring the majority class. Consequently, accuracy should be interpreted alongside class-sensitive metrics such as f1-score.

4.1.2 F1-score

The F1-score provides a harmonic mean between precision and recall, serving as a balanced indicator of performance when both metrics are important. It is defined as:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{16}$$

F1-score, is an informative measure that summarizes the trade-off between false positives (FP) and false negatives (FN). In practice, the F1-score is particularly useful for evaluating models under class imbalance, where accuracy may fail to reflect true discriminative performance. Unlike the arithmetic mean, the harmonic mean penalizes large disparities between precision and recall.

4.1.3 Precision

Precision evaluates the reliability of the model's positive predictions. It measures the proportion of correctly identified positive instances among all samples predicted as positive:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{17}$$

Parameter	Value
Train-test split ratio (%)	80 - 20
Training epochs	30
Batch size	8
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	3×10^{-4}
Dropout rate	0.2
Loss function	BCEWithLogitsLoss

Table 3: Parameter settings of the *Hyper Persona* (proposed method) on the Essays dataset.

High precision indicates that when the model predicts the positive class, it is usually correct despite producing few false positives. This property makes precision especially valuable in contexts where false alarms carry a high cost. Nevertheless, precision alone does not account for missed positive instances, which are instead captured by recall.

4.1.4 Recall

Recall, also known as sensitivity or the true positive rate, measures the model's ability to correctly identify all positive instances. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{18}$$

A high recall value indicates that the model successfully detects most of the true positives, with few false negatives. Recall is particularly critical in applications where missing a positive instance is more costly than incorrectly predicting one.

4.2 Parameter Settings

This section details the hyperparameters and implementation settings used in the experiments. As presented in Figure 3, the preprocessong 3.2.1 and text segmentation 3.2.2.1 were performed using spaCy [55]. Hierarchical graph 3.2.2.4 and hypergraph 3.2.2.3 structures were constructed with NetworkX [56] and HyperNetX [57], respectively. Primary features were represented using pre-computed embeddings extracted from BERT [14]. As illustrated in Figure 5, the proposed Transformer-Based Graph Encoder 3.2.3.3 was implemented using PyTorch [58] and PyTorch Geometric [59], We used two TransformerConv layers to project node features into a 128 and 64 dimensional latent space, followed by layer normalization and residual linear projection. At the end we performed global additive pooling to derive graph-level embedding in a 64 dimentional feature space.

As demonstrated in Figure 6, the Personality Classfier 3.2.4, is a classification head consisting of a linear layer (64→16), layer normalization, sigmoid activation, dropout, and a final linear layer to produce the output logits. During training, a learnable feature-selection mask was optimized using the Gumbel-Softmax 3.2.3.2 trick with a temperature of 0.5, enabling differentiable selection of informative features. All The experiments was performed using an NVIDIA GeForce RTX 4060 GPU with 8 GB of G-RAM.

4.3 Baseline Methods

In order to assess the effectiveness of the *Hyper Persona*, it is crucial to benchmark its performance against well-established baseline methods in personality prediction from text. These baselines represent a spectrum of methodological approaches, ranging from traditional machine learning and ontological reasoning to modern deep learning architectures leveraging contextual embeddings. Each method provides unique insights into how textual features can be harnessed for personality inference and therefore serves as a meaningful point of comparison.

- Tighe et al. [60] employed LIWC features to conduct APP on the Essays dataset using several classifiers, including SVM, Sequential Minimal Optimization, and Linear Logistic Regression. They focused on enhancing classification performance by eliminating insignificant LIWC features through Information Gain and Principal Component Analysis (PCA).

- Majumder et al. [13] proposed a convolutional neural network architecture to infer an author's Big Five personality traits from text. They built one binary classifier per trait, all sharing a common architecture, in which sentence-level convolutional layers encode features that are then aggregated to document-level representations.

- Ramezani et al. [24] integrated lexical, semantic, and sequential levels. Their ensemble combined term-frequency vectorization, ontology-based and enriched ontology models, latent semantic analysis, and BiLSTM-based deep learning. These base models were stacked into a meta-model using a Hierarchical Attention Network, which performed reasoning across levels.

- Wang et al. [61] proposed a graph convolutional neural network that represents users and their textual information within a graph structure, with the resulting embeddings classified via a softmax layer.

- Kazameini et al. [38] combined contextual BERT embeddings with an ensemble of bagged Support Vector Machines (SVMs). Their approach preserved token-level

Personality Trait	**Accuracy(%)**	**F1(%)**	**Precision(%)**	**Recall(%)**
O	68.69	67.64	72.32	63.53
C	61.82	60.21	63.84	56.97
E	64.44	65.22	66.01	64.45
A	60.31	67.74	58.45	61.53
N	60.53	64.48	58.61	61.82
Average	**63.15**	**65.05**	**63.84**	**61.82**

Table 4: Hyper Persona (proposed method) evaluation results on Essays Dataset per trait

representations and aggregated them through bootstrap sampling, effectively bridging contextual embeddings with robust ensemble learning.

- El-Demerdash et al. [62] proposed the use of Universal Language Model Fine-Tuning (ULM-FiT) for author personality prediction (APP), an effective transfer learning approach applicable across various natural language processing tasks.
- Jiang et al. [63] leveraged pre-trained contextual embeddings, specifically BERT and RoBERTa, to enhance the accuracy of predictions in author personality prediction (APP).

4.4 Evaluation Results

Table 4 provides a detailed per-trait evaluation of our model across standard performance metrics. Openness exhibits the highest overall performance, indicating that the model is most effective in predicting this trait. Extraversion also yields strong and balanced results across all metrics, suggesting consistent performance. Conscientiousness, Agreeableness, and Neuroticism show slightly lower performance, reflecting the greater challenge of inferring these traits from text. The average accuracy across all traits is 63.15%, with an F1-score of 65.05%. Additionally, the model achieves 63.82% precision and 61.82% recall, showing its ability to make reliable and consistent predictions. These results demonstrate that *Hyper Persona* delivers stable and competitive performance across all traits, with particularly strong predictive capabilities.

Figure 7, presents a comparative visualization of the model's performance across the five personality traits underscoring the model's effectiveness in multi-trait prediction, with particularly strong generalization for cognitively and socially expressive traits.

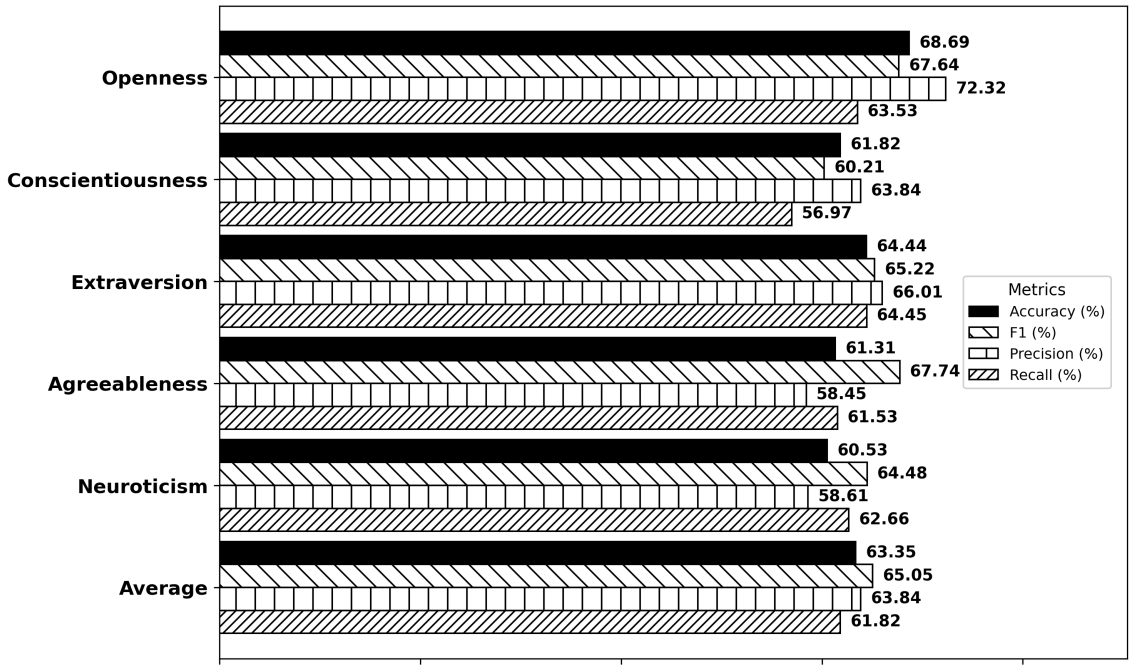


Figure 7: Performance of the proposed Hyper Persona model across the Big Five personality traits, evaluated using Accuracy, F1-score, Precision, and Recall metrics.

Table 5 presents a comparative analysis against baseline methods. Earlier approaches, such as those by Majumder et al. [13], El-Demerdash et al. [62], and Jiang et al. [63], achieved moderate performance levels, reflecting the inherent difficulty of modeling personality traits solely from textual data, particularly within the Essays dataset. Ontology-based and semantically enriched architectures, such as Ramezani et al. [24], achieved modest improvements by leveraging structured lexical and semantic information. In contrast, *Hyper Persona* surpasses all baseline methods across three personality traits and in overall performance, highlighting its ability to extract richer contextual cues from textual input. Although the model shows slightly lower results in agreeableness and neuroticism, this can be attributed to the subtler and less distinctive linguistic markers associated with these traits within the data set. The overall results demonstrate that *Hyper Persona* offers a substantial and stable advancement over existing methods, effectively capturing both expressive and cognitive language patterns.

Figure 8 illustrates the comparative classification accuracy across the Big Five personality dimensions, along with the overall average. The results confirm that the proposed *Hyper Persona* model consistently outperforms prior approaches, achieving the highest average accuracy. This consistent improvement highlights the model's capability to effectively capture intricate linguistic and contextual cues relevant to personality expression.

Baselines	Accuracy (%)					
	O	C	E	A	N	Avg.
Tighe et al. [60]	61.95	56.04	55.75	57.54	58.31	57.92
Kazamini et al. [38]	62.09	57.84	59.30	56.54	59.39	59.03
Ramezani et al. [24]	56.30	59.18	64.25	60.31	61.14	60.24
Wang et al. [61]	64.80	59.10	60.00	57.70	**63.00**	60.92
Jiang et al. [63]	65.86	58.55	60.62	59.72	61.04	61.16
Majumder et al. [13]	62.68	57.30	58.09	56.71	59.38	58.83
El-Demerdash et al. [62]	65.60	59.52	61.15	60.80	62.20	61.85
Hyper Persona (Ours)	**68.69**	**61.82**	**64.44**	**61.31**	60.53	**63.35**

Table 5: Comparison of Accuracy across baselines.

Text Level(s)	Accuracy (%)					
	O	C	E	A	N	Avg.
Multi-level (Proposed)	**68.69**	**61.82**	**64.44**	**61.31**	**60.53**	**63.35**
Document-level + Sentence-level	67.81	59.51	62.35	59.11	59.31	61.61
Document-level + Word-level	66.80	58.88	61.15	58.10	58.97	60.66
Sentence-level	64.19	55.31	58.55	57.16	57.12	58.46
Word-level	63.82	55.08	57.22	56.73	56.94	57.95

Table 6: Ablation study across different representation level combiantion for each Big Five personality trait.

4.5 Discussion

This study was designed to assess the effect of multi-level text representation using hypergraph and hierarchical graph structures in automatic personality prediction (APP). To this end, we proposed *Hyper Persona*, a framework that leverages both hypergraph and hierarchical graph structures to represent the multi-level organization of written text (document, sentence, and word) and employs a transformer-based graph encoder for feature learning.

Addressing **RQ1**, the proposed multi-level hypergraph structure proves crucial for enriching the semantic and syntactic representation of text. By jointly modeling word, sentence, and document-level interactions, the framework captures both local lexical associations and global contextual dependencies. Unlike traditional flat or sequential representations, it encodes relationships among linguistic units at multiple granularities, enabling the model to preserve discourse coherence while retaining fine-grained syntactic and semantic nuances. This integration yields a unified representation that bridges micro-level lexical patterns and macro-level thematic organization, providing a richer foundation for personality inference.

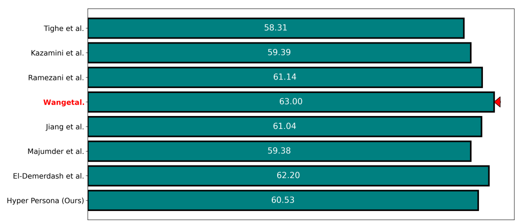
Moving to **RQ2**, the results presented in Figure 7 and Table 4 demonstrate that the transformer operating over the hypergraph effectively aggregates personality-relevant linguistic signals across multiple levels of abstraction. Through the attention mechanism,

A Openness

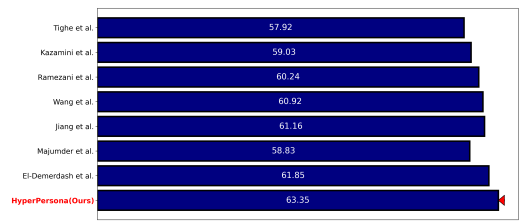
B Conscientiousness

C Extraversion

D Agreeableness



E Neuroticism



F Average

Figure 8: Comparative accuracy results per personality trait (A, B, C, D and E) and overall average (F).

the model selectively emphasizes high-impact nodes, such as semantically salient words or sentences that convey personality cues (for instance, emotional intensity for Neuroticism or social expressiveness for Extraversion). This hierarchical propagation enhances both the interpretability and the sensitivity of the learned embeddings, showing that the hypergraph-transformer architecture generalizes personality indicators across different linguistic layers while maintaining balanced performance across all traits.

In relation to **RQ3**, the comparative results illustrated in Figure 8 and Table 5 reveal that the proposed *Hyper Persona* framework consistently surpasses established baseline models across most of the Big Five traits. The model's capacity to jointly encode multi-level structural dependencies enables it to extract richer personality cues from text than single-level or sequential approaches. Its superiority is particularly evident for Openness, Conscientiousness, Extraversion, and Agreeableness, where nuanced linguistic variety and social tone are more effectively captured through hierarchical modeling. Even for traits such as Neuroticism, where personality expressions are linguistically subtle, the model maintains

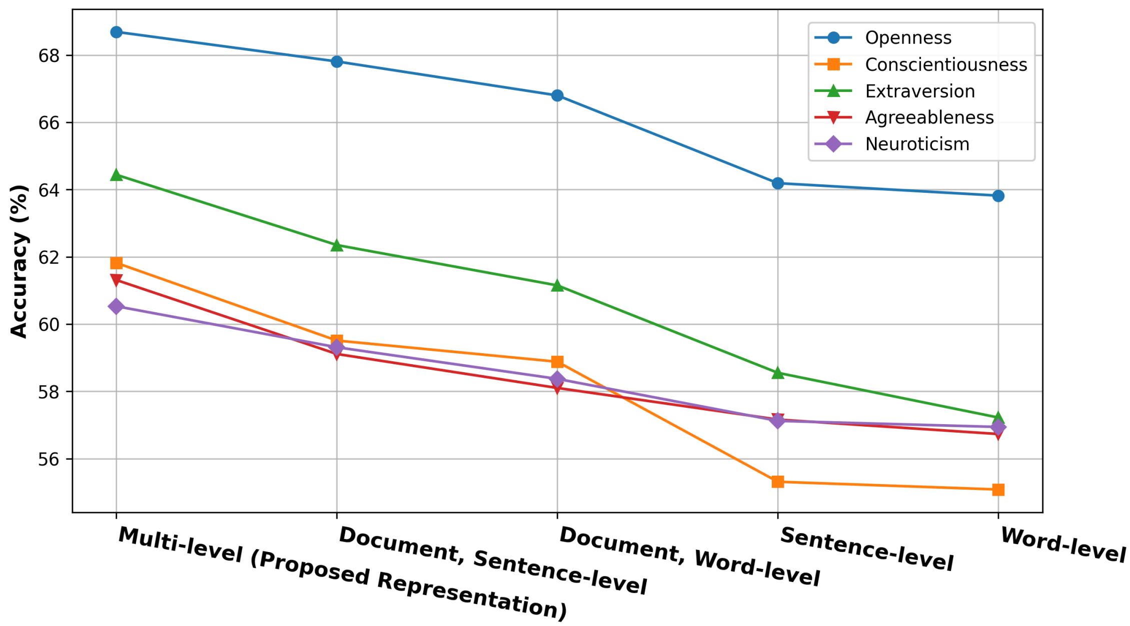


Figure 9: Ablation study across different representation level combiantion for each Big Five personality trait.

stable and competitive performance highlighting its robustness in handling diverse textual styles and personality manifestations.

Finally, addressing **RQ4**, Figure 9 and Table 6 reveal a clear performance decline as the representation shifts from the full multi-level structure to simpler single-level variants. This trend quantifies the contribution of each abstraction layer: document-level features provide global thematic context; sentence-level connections encode syntactic and discourse structure; and word-level cues capture lexical and affective signals. The synergy among these layers yields the most accurate personality representations. In contrast, when individual levels are isolated (such as in sentence or word-only configurations) the model loses access to important contextual dependencies, leading to notable performance degradation. These results affirm the necessity of multi-granular fusion for robust and comprehensive personality inference.

5 Conclusion

This study was designed to examine the impact of multi-level text representations using hypergraphs and hierarchical graph structures, and transformer-based graph encoders on text-based automatic personality prediction (APP). To this end, we proposed *Hyper*

Persona, a framework that leverages hypergraph and hierarchical graph structures to capture the multi-level nature of written text (document, sentence and word) and employs a transformer-based graph encoder for feature learning. *Hyper persona* operates through four main phases. First, a preprocessing phase prepares the input text for subsequent processing. Second, a text representation strategy transforms each sample to a digestable form for machine. Third, representation learning and dimensionality reduction are performed using a transformer-based graph encoder. Finally, learned representations are classified using a multi-layer perceptron.To the best of our knowledge, this is the first work to employ hypergraph and hierarchical graph structures for text representation to perform text-based APP. The findings from this study offer several noteworthy contributions to the existing literature. First, the proposed multi-level approach significantly improves the accuracy of text-based APP systems that rely solely on written text. Second, embedding text using pre-trained transformers provides contextual awareness and eliminates the need for traditional preprocessing techniques. Third, this study advances our understanding of how hypergraphs and hierarchical graphs can effectively exploit textual hierarchy, offering a more comprehensive and structured representation of written text compared to conventional flat sequences of characters highlighting the importance of the proposed multi-level representation learning in enabling machines to perform more human-like decision-making.

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